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26/08/26 Computer Vision

Application Problems:

1) High-boost filtering: Implement a system that enhances high frequency components while retaining part of the original image: $f_{hb} = (A-1) f_{orig} + f_{sharpened}$

Any Given: $f_{hb} = (A-1) f_{orig} + f_{sharpened}$

Suppose: Original pixel value: $f_{orig} = 100$
Sharpened pixel value: $f_{sharpened} = 180$
$A = 2$

Substituting values: $f_{hb} = (2-1)(100) + 180$

$f_{hb} = 1(100) + 180$

$f_{hb} = \underline{\underline{280}}$

∴ The resulting high-boost filtered pixel value is 280.
High-boost Filtering enhances the high-frequency details such as edges and fine features while still retaining part of the original image.

2.) Adaptive Local Enhancement using Gamma Correction

Ans The Gamma correction formula is:

$$s = r^{\gamma}$$

where r = normalised input intensity

γ = gamma value

s = output intensity

For a dark region suppose:

$r = 0.25$

$\gamma = 0.5$

Then $s = r^{\gamma}$

$= (0.25)^{0.5}$

$s = \underline{\underline{0.5}}$

∴ So, the dark pixel changes from 0.25 → 0.5 making the region brighter.

For a brighter region:

Suppose: $r = 0.81$, $\gamma = 2$

Then: $s = (0.81)^2$

$s = 0.6561$

So the intensity changes from 0.81 → 0.6561, reducing the brightness.